Questions by topic: Renewable energy technologies - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the efficiency and cost of four types of solar photovoltaic (PV) cell.

Table 1

PV cell type	Efficiency / %	Relative cost per m^2
Amorphous silicon (thin film)	8	1
Polycrystalline silicon	16	2
Monocrystalline silicon with anti-reflective surface	22	3
Multi-junction cell	40	25

0 1 . 1 Describe how a photovoltaic cell generates electricity.

[2 marks]

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0 1 . 2 Explain how anti-reflective surfaces and multi-junction cells increase efficiency.

[3 marks]

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0 1 . 3 Use Table 1 to suggest which cell type would be chosen for a large solar farm on cheap desert land and which for a satellite. Justify your choices.

[3 marks]

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0 1 . 4 Explain why solar PV output is intermittent **and** unpredictable.

[2 marks]

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0 2 . 1 Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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0 2 . 2 Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

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0 2 . 3 Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

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0 2 . 4 Describe **two** environmental impacts of solar PV.

[2 marks]

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0 3 A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

0 3 . 1 Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

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0 3 . 2 Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

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0 3 . 3 The maximum flow rate at the site is 6.5 m^3 s^{-1} giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

Answer ..... kW

0 3 . 4 Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

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0 4 The kinetic energy available to a wind turbine is proportional to the cube of the wind velocity. Table 1 shows mean wind velocities at four possible wind farm sites.

Table 1

Site	Mean wind velocity / m s^{-1}	Distance to grid / km	Other features
A: upland ridge	9	12	Migrating bird route; peat bog
B: coastal farmland	8	3	Near village
C: offshore shallow sea	10	15	Sandbanks
D: inland plain	6	2	Arable farmland

0 4 . 1 Calculate how many times more kinetic energy is available at site C than at site D.

[2 marks]

Answer ☐ times

0 4 . 2 Use Table 1 to evaluate the four sites for a large wind farm.

[5 marks]

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0 4 . 3 Give **two** advantages of vertical axis wind turbines (VAWTs) over horizontal axis turbines (HAWTs).

[2 marks]

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0 4 . 4 Explain how direct-drive turbines improve on turbines with a gearbox.

[1 mark]

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0 5 . 1 Explain the locational requirements for harnessing wave power.

[2 marks]

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0 5 . 2 Describe how **one** named wave power device generates electricity.

[2 marks]

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0 5 . 3 Give **two** design problems that have delayed the development of wave power.

[2 marks]

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0 5 . 4 Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

[4 marks]

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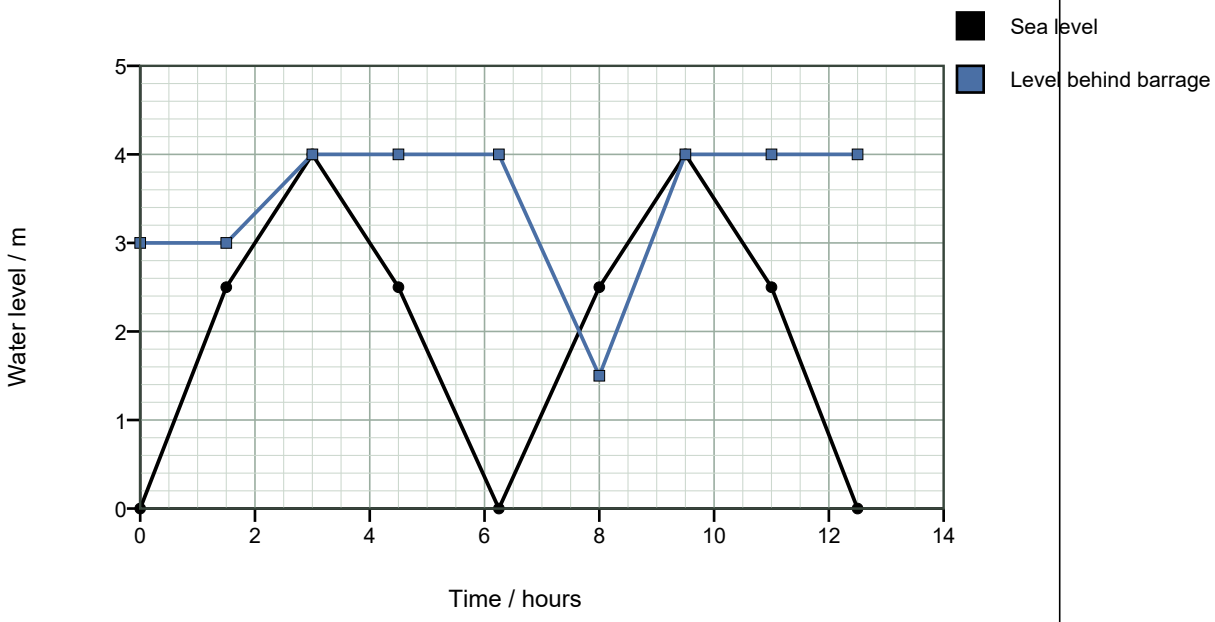
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0 6 Figure 1 shows the water level inside and outside a tidal barrage over one 12.5-hour tidal cycle and the periods when electricity is generated.

Figure 1



0 6 . 1 Use Figure 1 to explain when electricity is generated by the barrage.

[2 marks]

0 6 . 2 Explain why tidal power is intermittent but predictable.

[2 marks]

0 6 . 3 Explain the difference between spring and neap tides and why this affects output.

[2 marks]

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0 6 . 4 Explain how a tidal lagoon can be used to generate electricity when there is no tidal flow.

[2 marks]

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0 6 . 5 Give **two** coastal features that make a site suitable for tidal power.

[2 marks]

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0 7 . 1 Explain **three** advantages of biofuels over other renewable energy resources.

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0 7 . 2 Explain **three** disadvantages of biofuel crops.

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0 7 . 3 Describe **two** new biofuel technologies.

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0 7 . 4 Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

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0 8 . 1 Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

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0 8 . 2 State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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END OF QUESTIONS